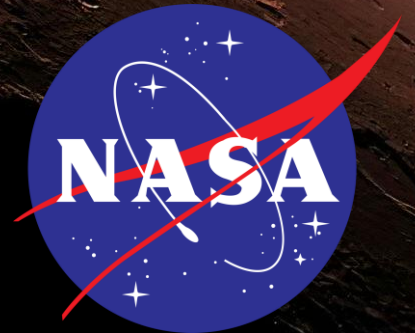


Mars Airfield Evaluation and Survey Tool for Rotorcraft Operations (MAESTRO)



Visud Chang, UC Berkeley
Mission Design Center, NASA Ames



What is MAESTRO?

Rotorcraft provide several advantages over rovers in the exploration of the Martian surface, offering faster traversal and can access otherwise unreachable terrain, making them ideal for scouting and science missions.

As rotorcraft technology improves, new tools are needed for landing zone selection and path planning, such as MAESTRO. By identifying safe airfields and visualizing terrain constraints, MAESTRO helps reduce risks and open new opportunities for scientific exploration on future Mars rotorcraft missions.



Fig. 1: Mars Ingenuity



Fig. 2: Mars Chopper

What does MAESTRO do?

- ◆ Analyzes Mars topography using High Resolution Imaging Science Experiment (HiRISE) stereo-derived digital terrain models (1 m/pixel resolution, x, y, & z)
- ◆ Evaluates terrain to generate safe traversal plans and identify viable landing zones
- ◆ Allows users to define:
 - ◆ A region of interest
 - ◆ Safe airfield criteria (e.g., 10 m × 10 m area with $<10^\circ$ average slope)
 - ◆ Elevation scaling factor to highlight terrain features
- ◆ Automatically detects and marks safe landing zones (shown in green)
- ◆ Plots topographic elevation profiles between any two points
- ◆ Highlights safe airfields along planned paths for both nominal and emergency landing planning

HiRISE Imagery

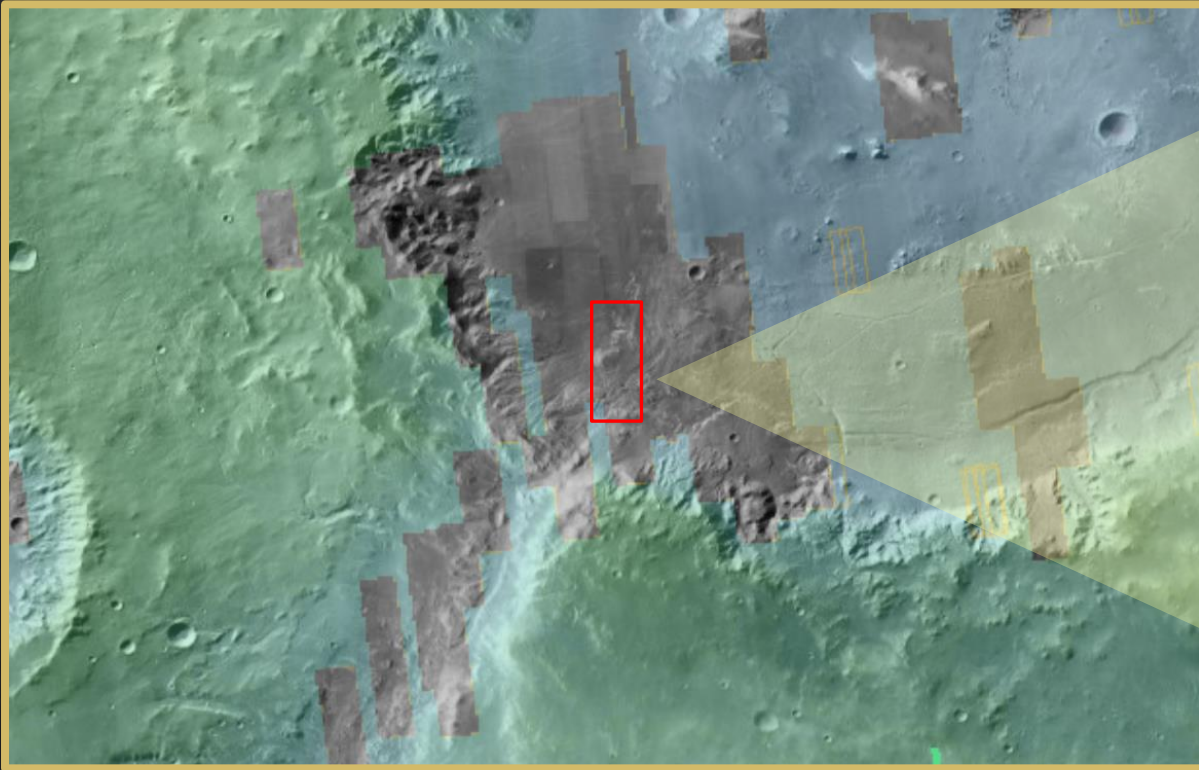
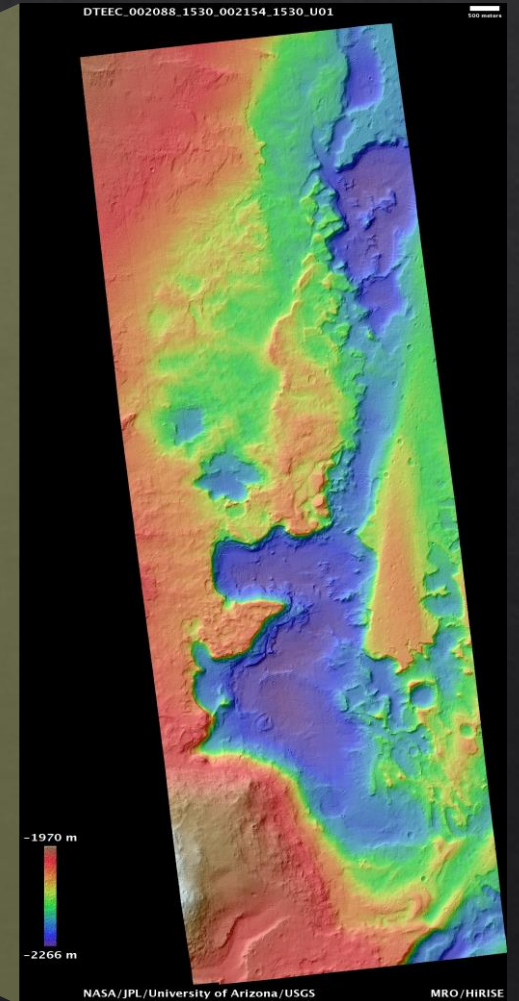
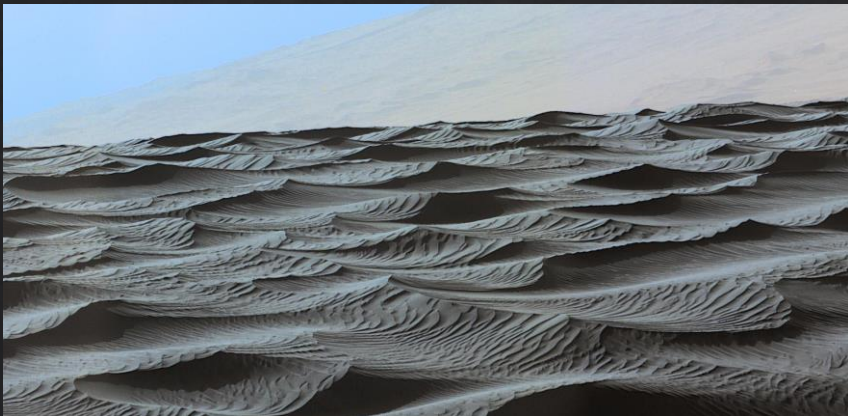
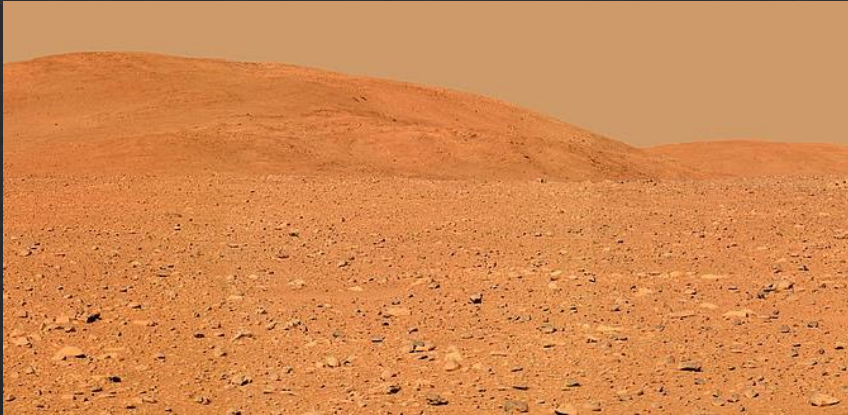


Fig. 3: Fan Surfaces in West Holden Crater
26.6°S, 34.8°W

HiRISE images have a resolution of 25 cm/pixel, but when combined into DTMs, the resolution becomes 1 m/pixel.





Types of Terrain

Since HiRISE imagery isn't available everywhere, similar mapped terrain can be used as a proxy to guide initial mission design and route planning. Analyzing these features can provide insight into the challenges and constraints mission operations planners must take into consideration when traversing various terrain.

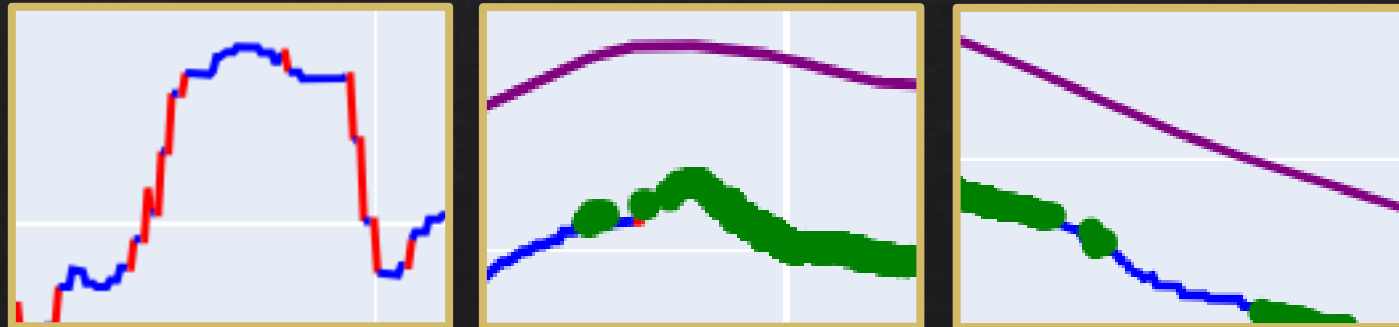
- Brain Terrain
- Flat Plains
- Dunes/Ridges
- Inclines
- Large Craters
- Long Traversals

MAESTRO's Topographic Profiles

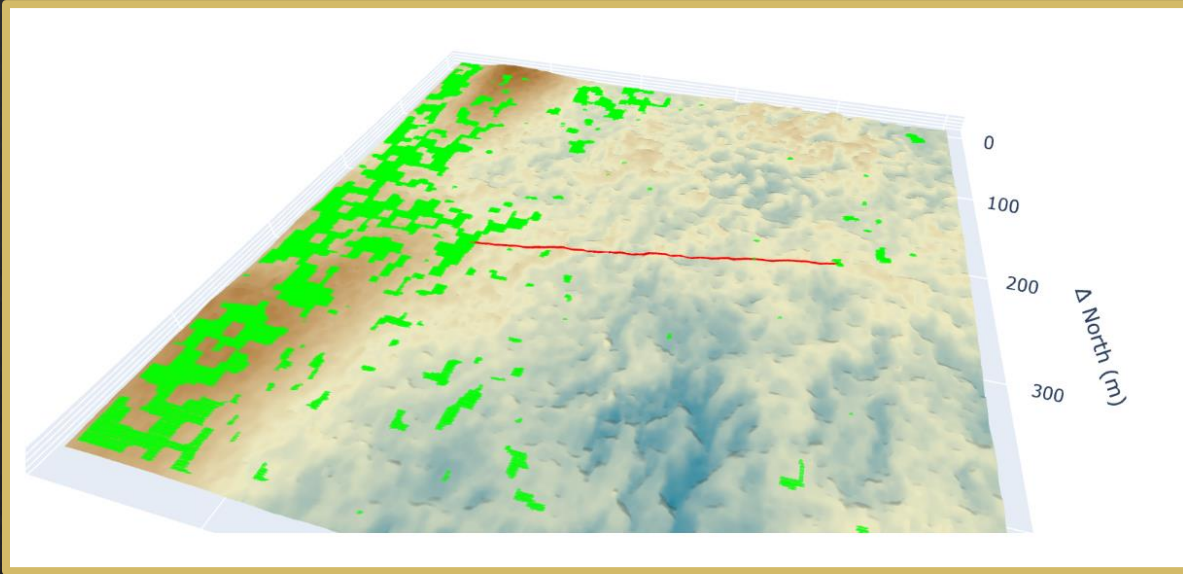
The topographic profiles of four common types of Martian terrain are shown with:

- ◊ **Red**: steep zones ($>15^\circ$ slope) to avoid
- ◊ **Green circles**: viable 10 m x 10 m landing zones within 50 m of the flight path
- ◊ **Purple**: a possible GPR-enabled flight path, offset 4 m above terrain with smoothing applied

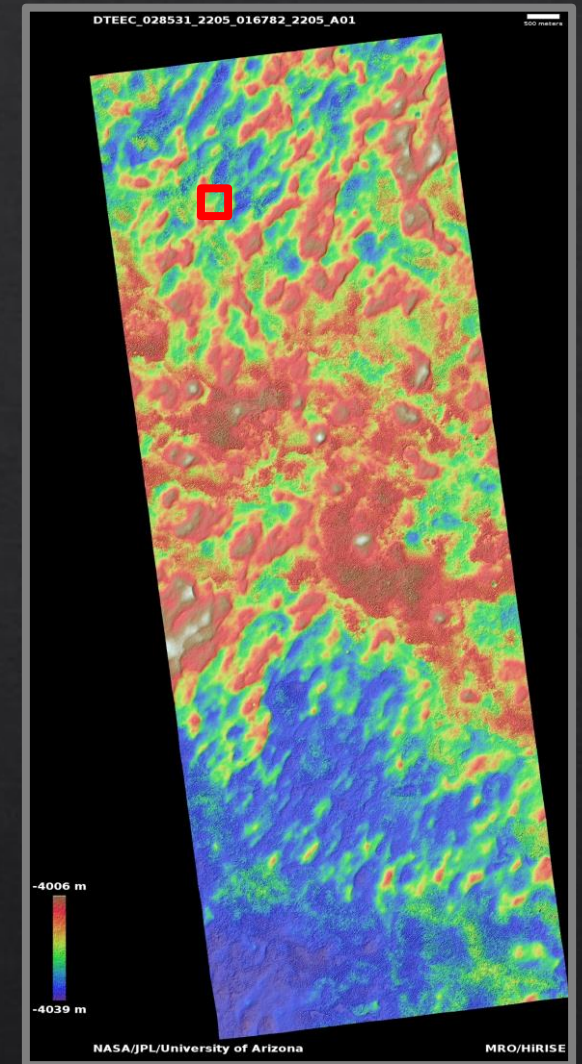
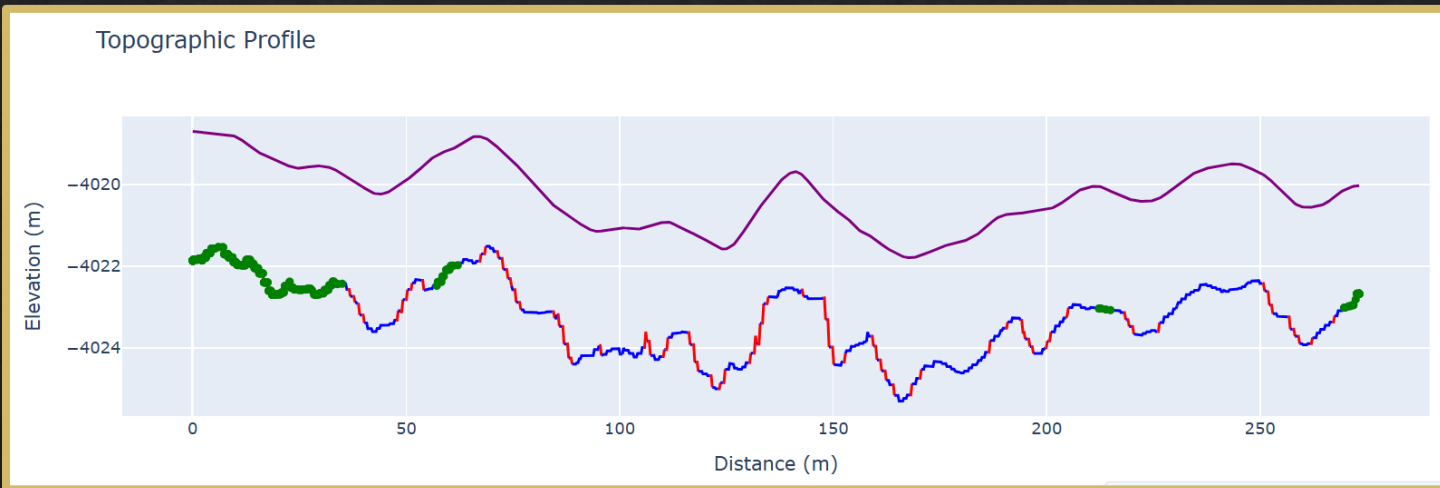
By highlighting backup airfields and steeper slopes along a route, MAESTRO helps mission planners make more informed decisions about flight paths.



Brain Terrain

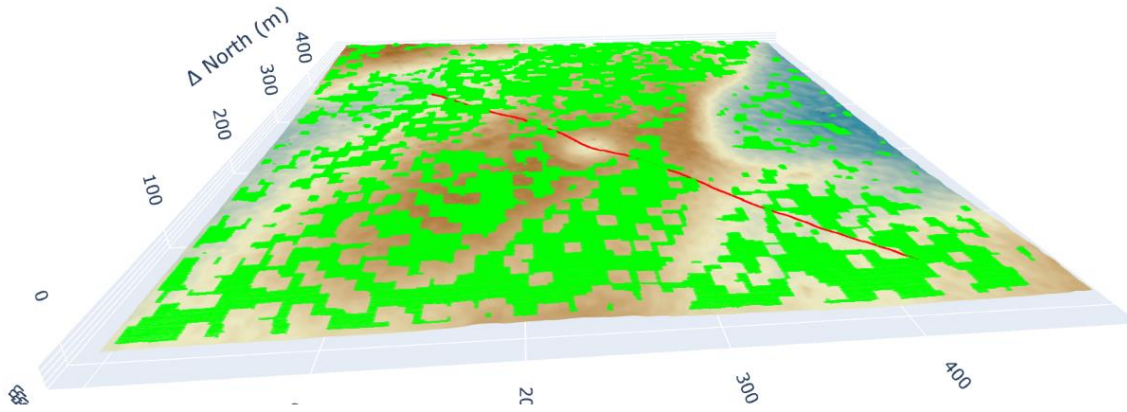


- Few viable landing zones
- Route planning is critical
- Precise altitude control required to maintain 4m offset above ground



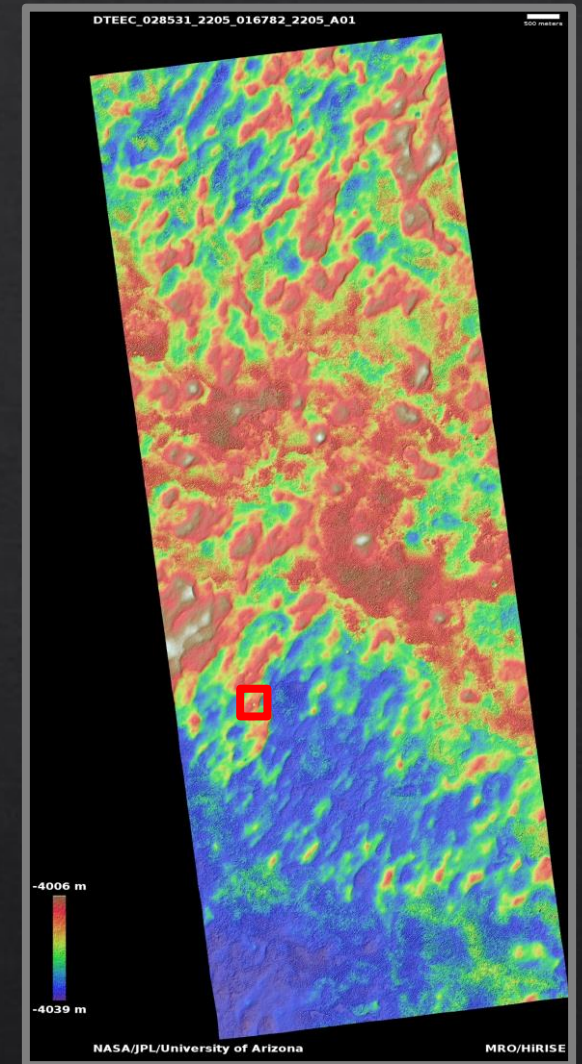
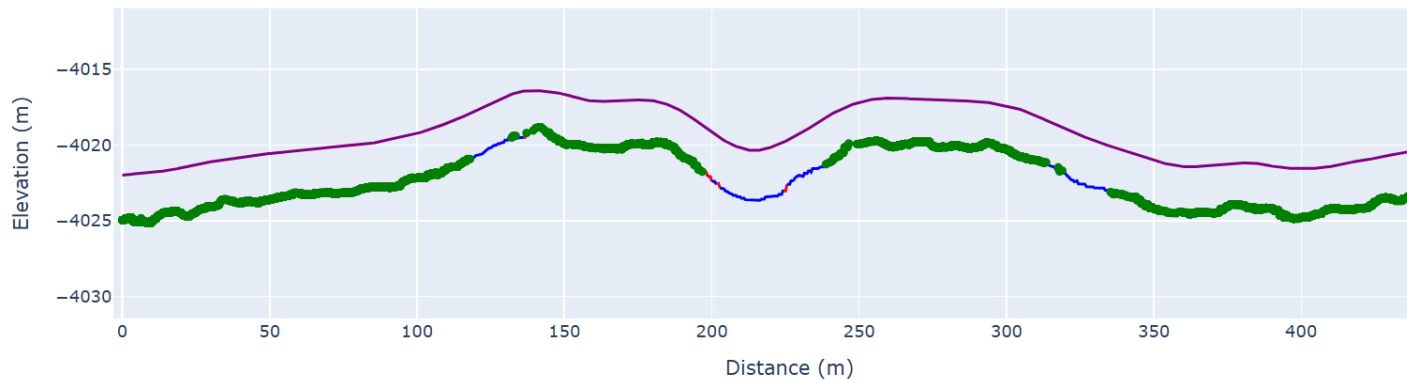
Northern Plains West of Erebus Montes
40.1°N, 177.6°W

Flat Plains



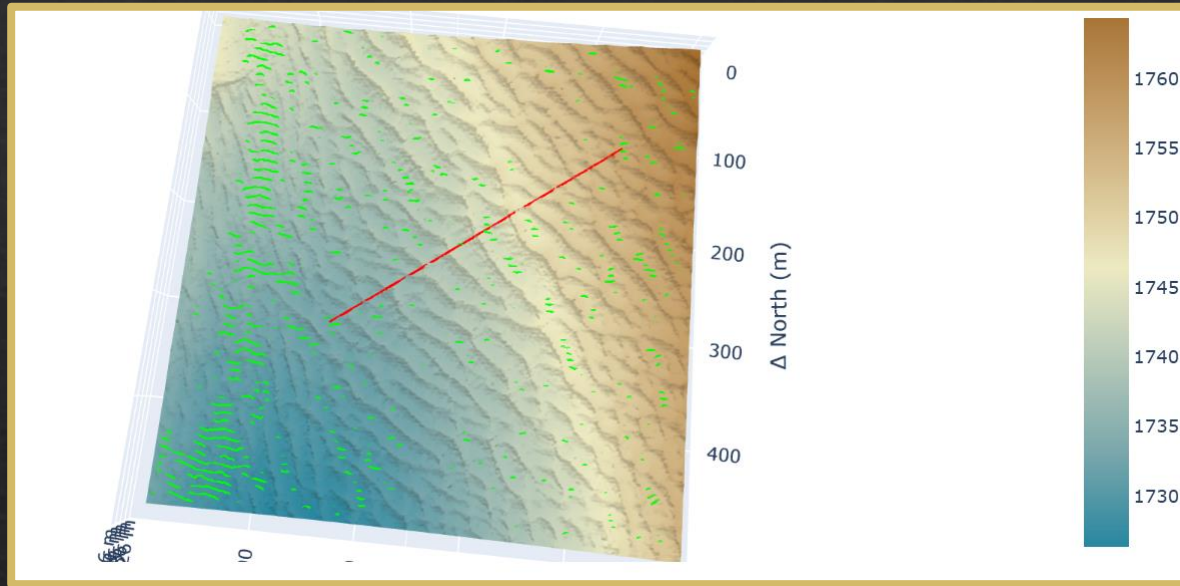
- Dense airfield coverage enables flexible routing.
- No airfields in crater.
- Ideal for extended traversals.

Topographic Profile

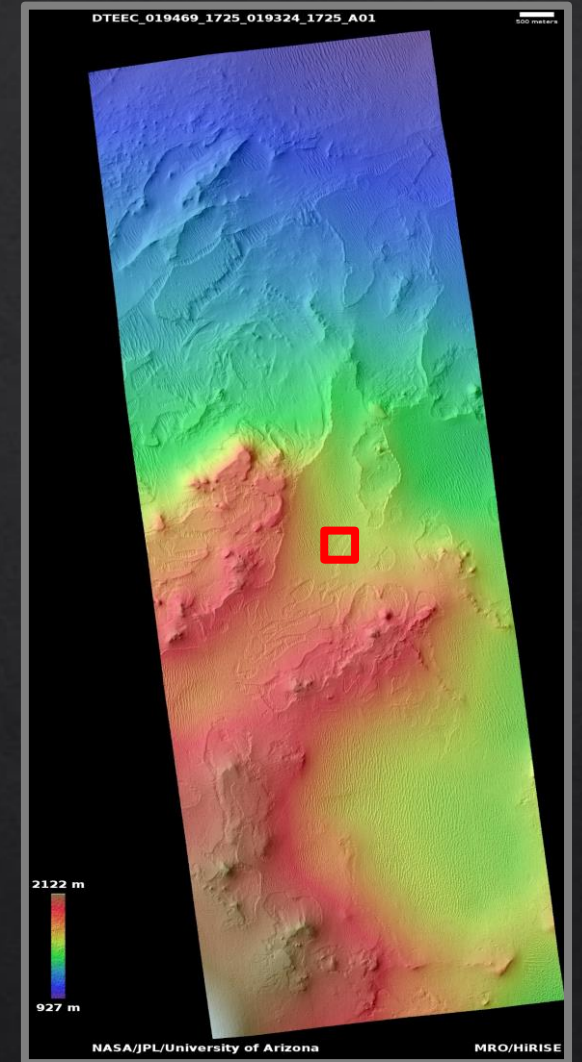
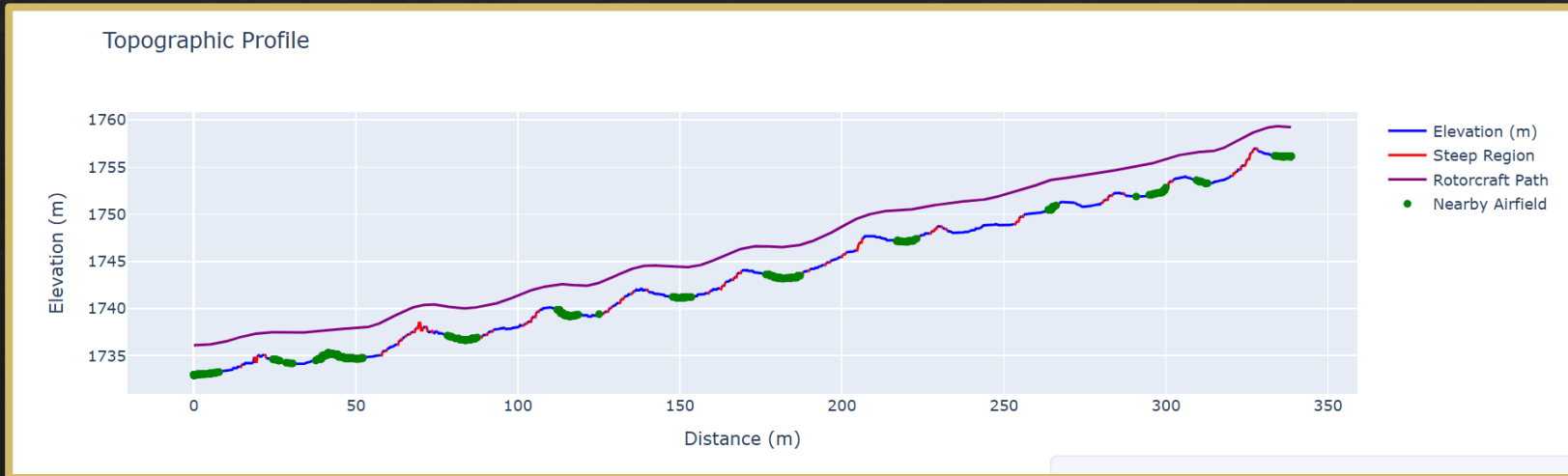


Northern Plains West of Erebus Montes
40.1°N, 177.6°W

Dunes/Ridges

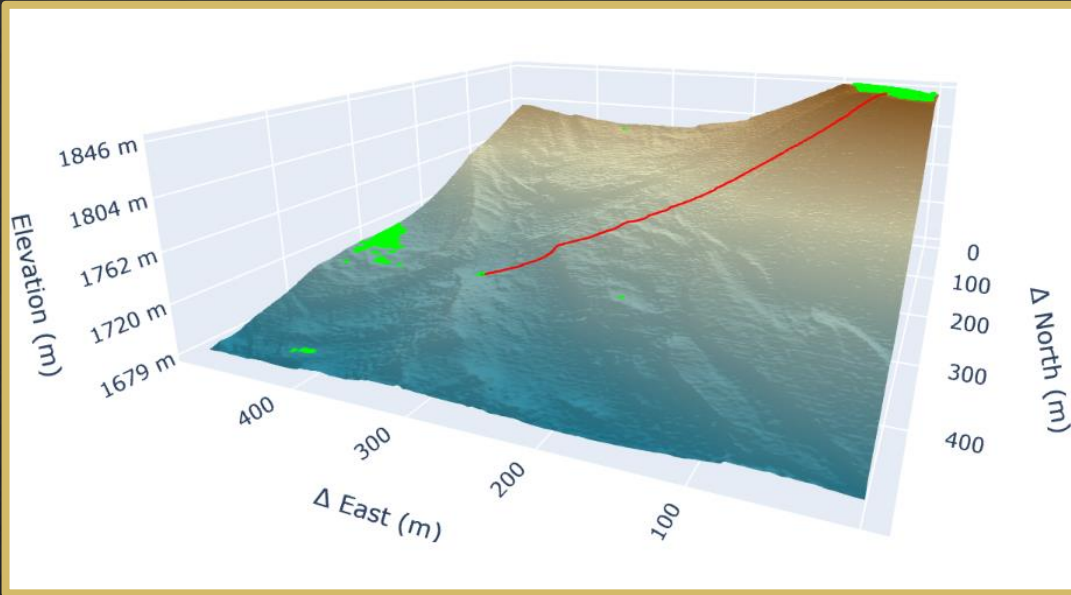


- Troughs offer sparse but regular landing sites $\sim 30\text{--}40$ m apart.

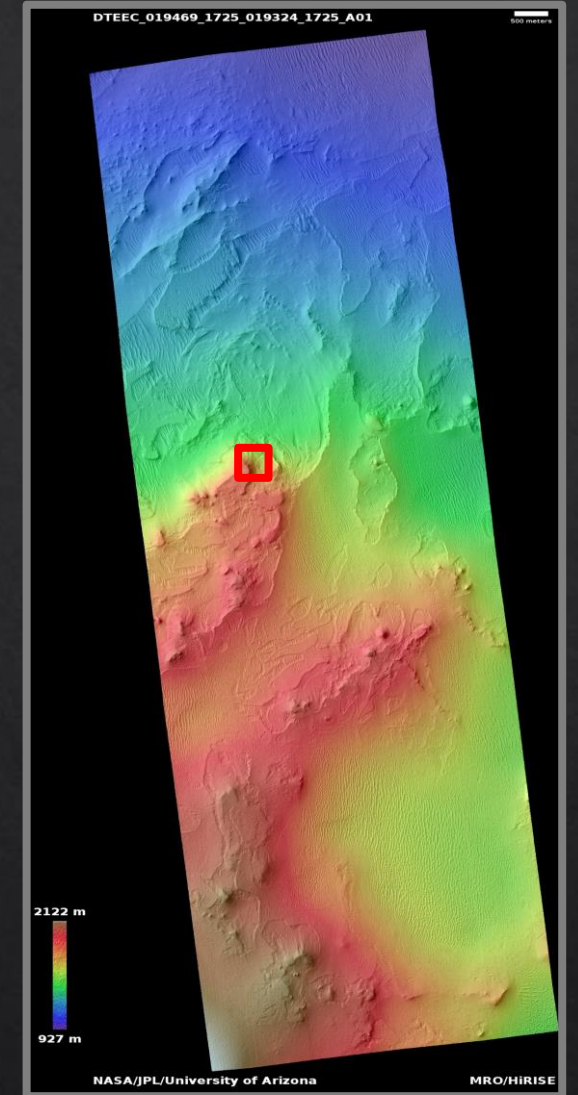
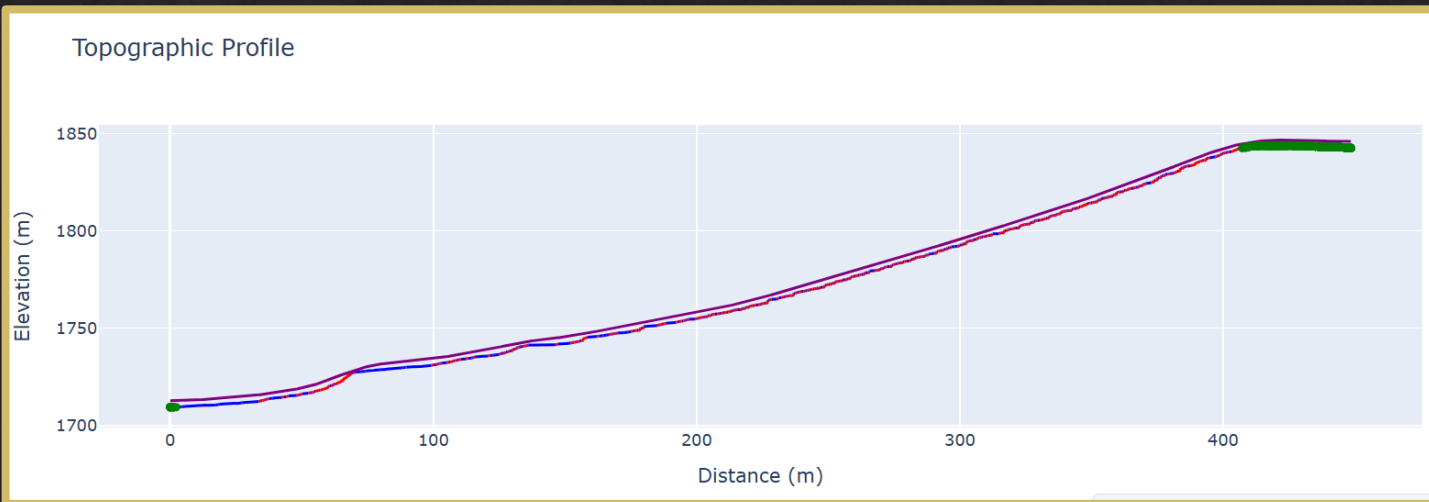


Light-Toned Outcrops in Noctis
Labyrinthus Region
 13.4°S , 96.6°W

Incline

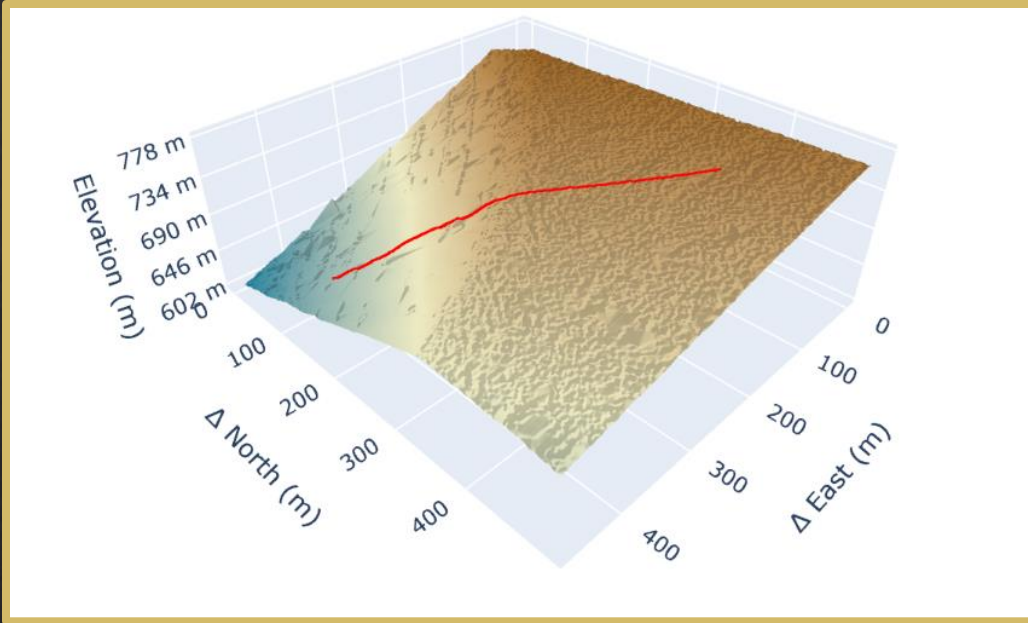


- Long slope segments without safe landing zones require sustained flight and backup plans.

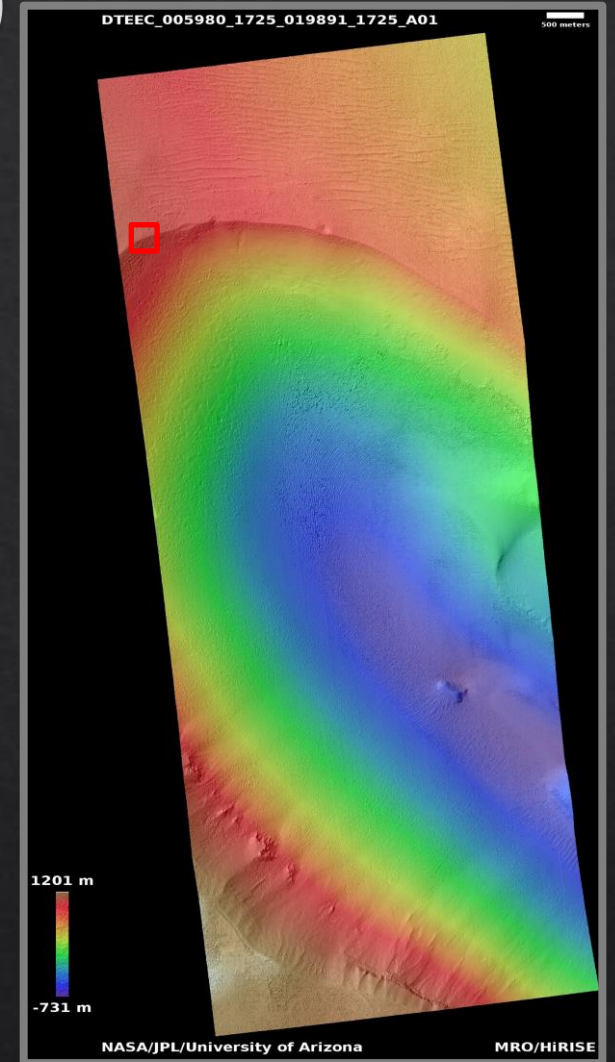
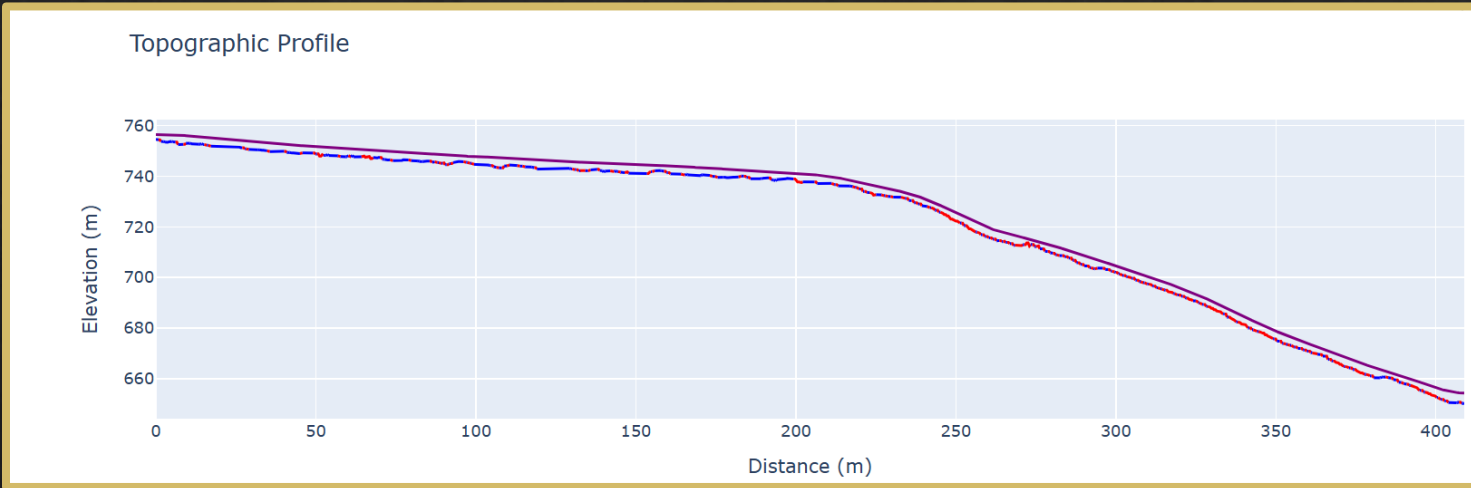


Light-Toned Outcrops in Noctis
Labyrinthus Region
13.4°S, 96.6°W

Large Crater (No Landing Sites)

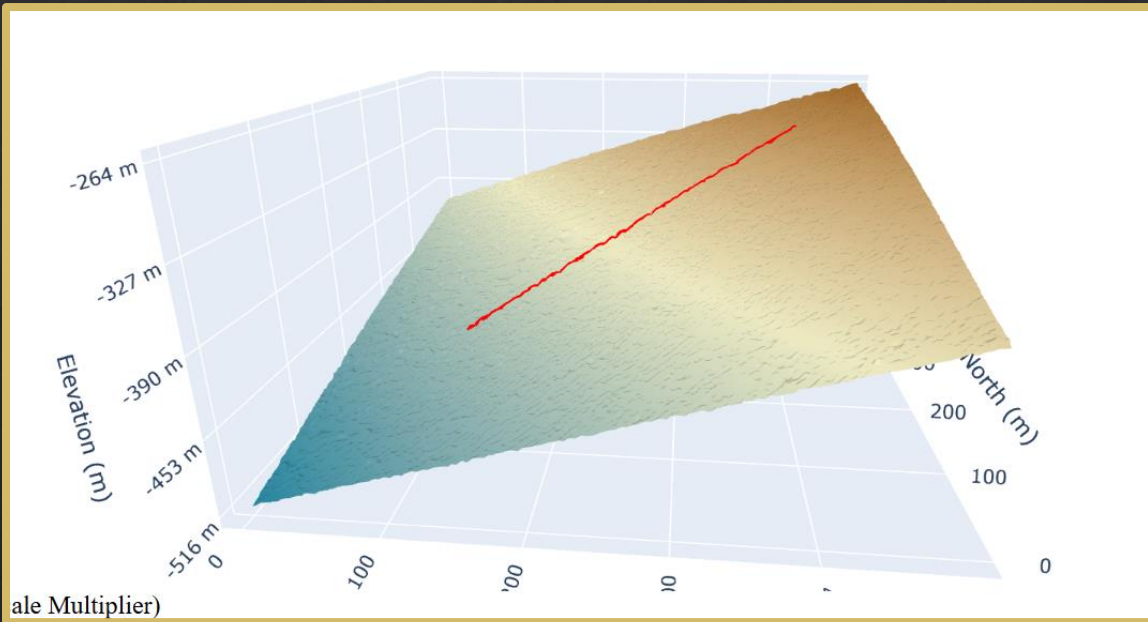


- No airfields detected in this region.
- Northwest terrain is too rocky; southeast terrain is too steep.
- Must approach crater floor from another direction.

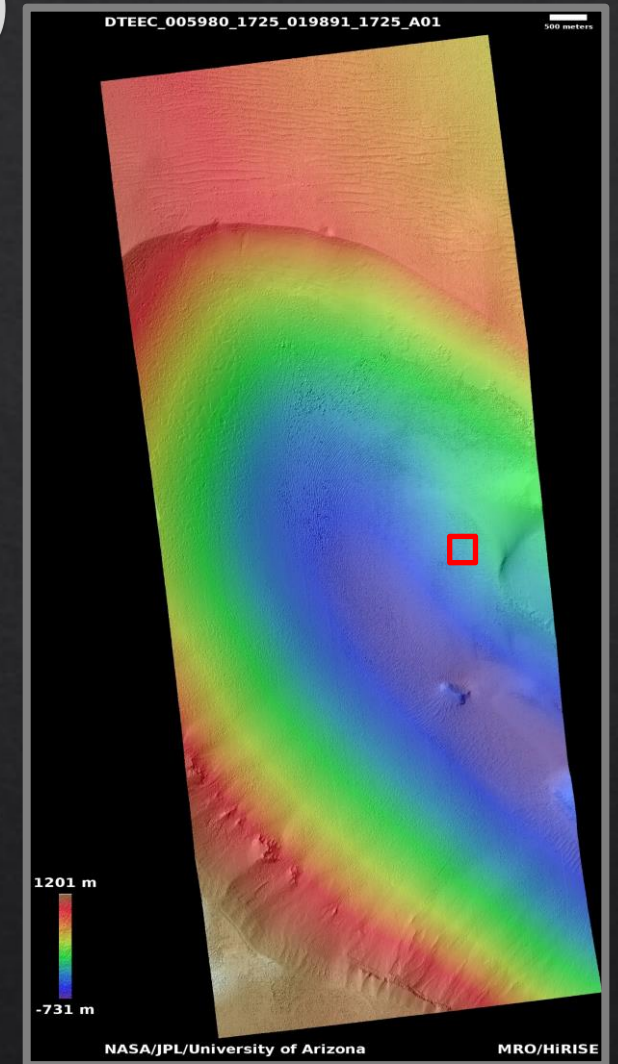
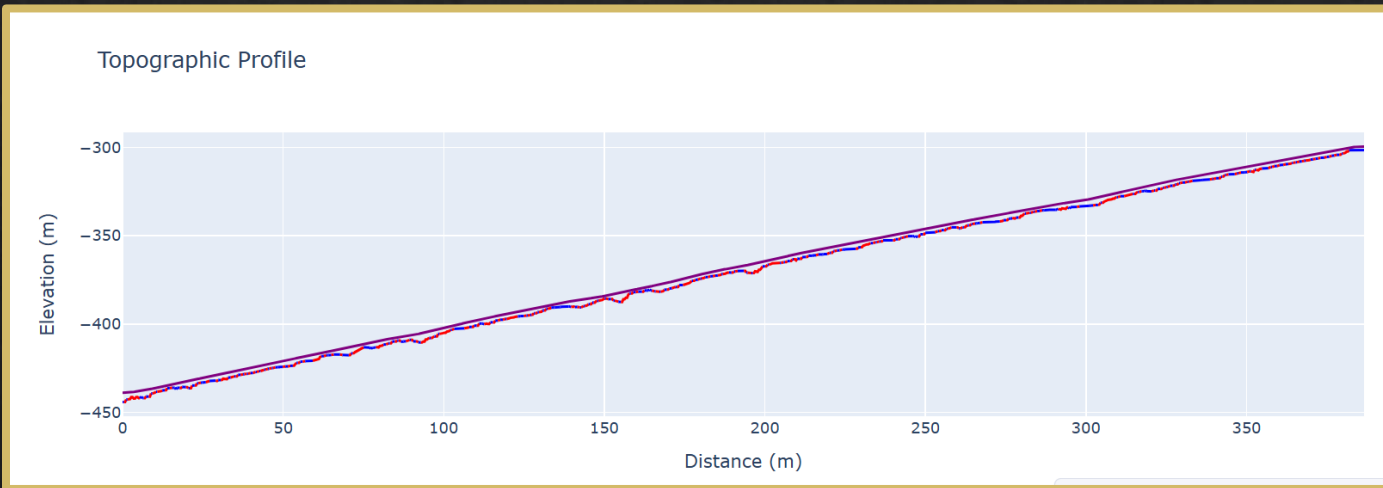


Layered Pit in Noctis Labyrinthus
7.7°S, 92.8°W

Large Crater (No Landing Sites)

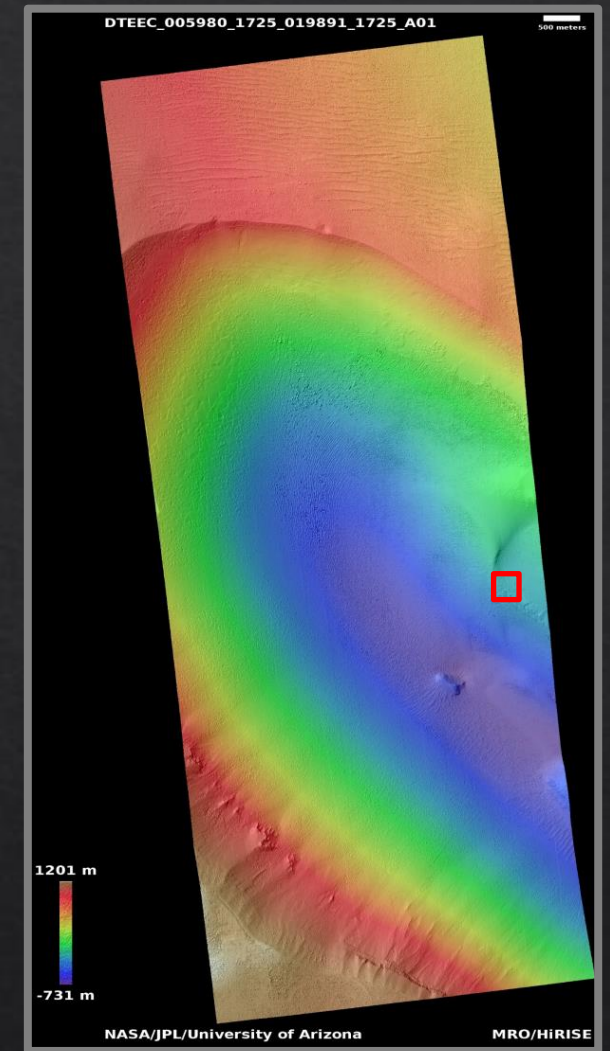
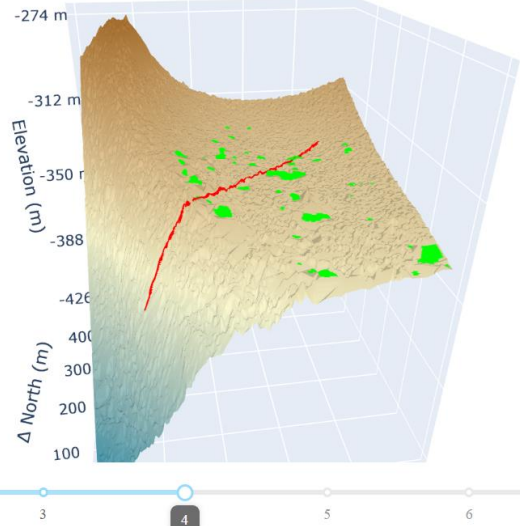
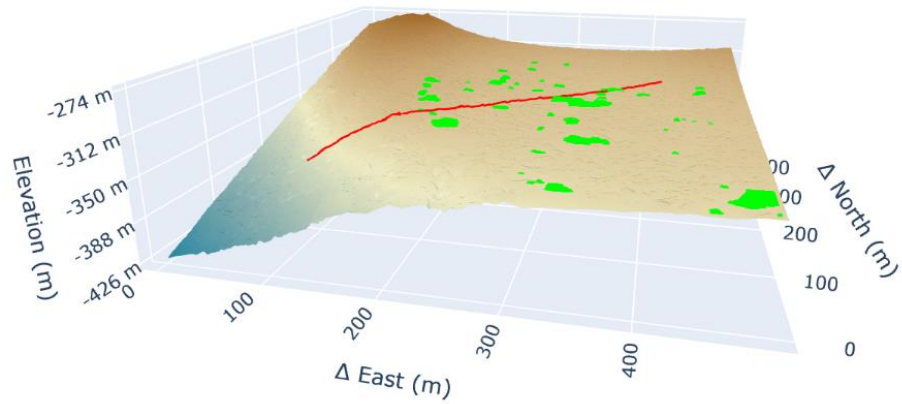


- Continuous steep slopes across the segment leave no safe landing zones.
- Emergency landing is infeasible—entire route requires sustained flight.

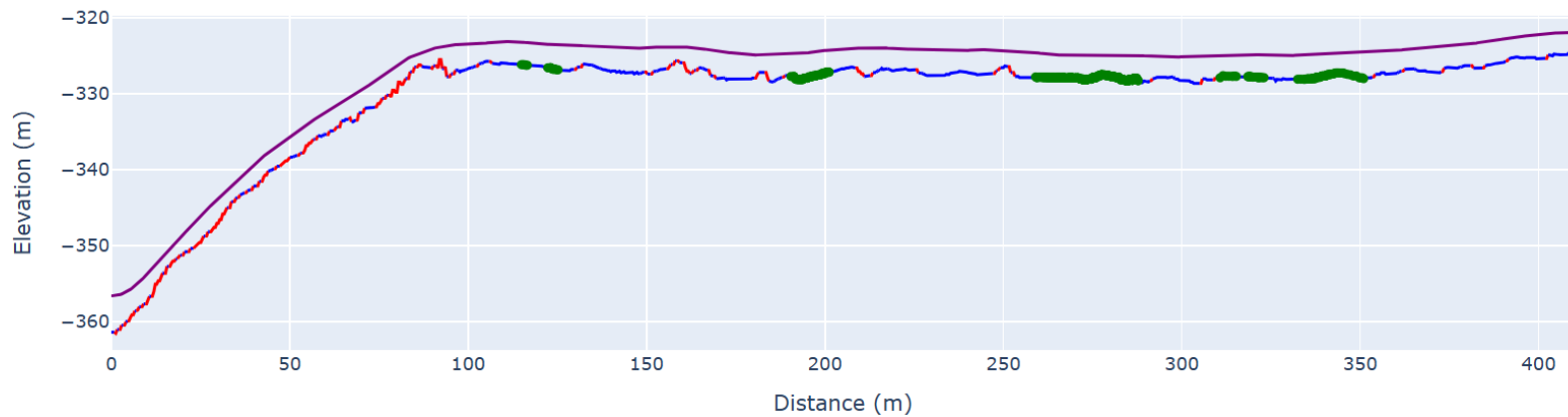


Layered Pit in Noctis Labyrinthus
7.7°S, 92.8°W

Large Crater

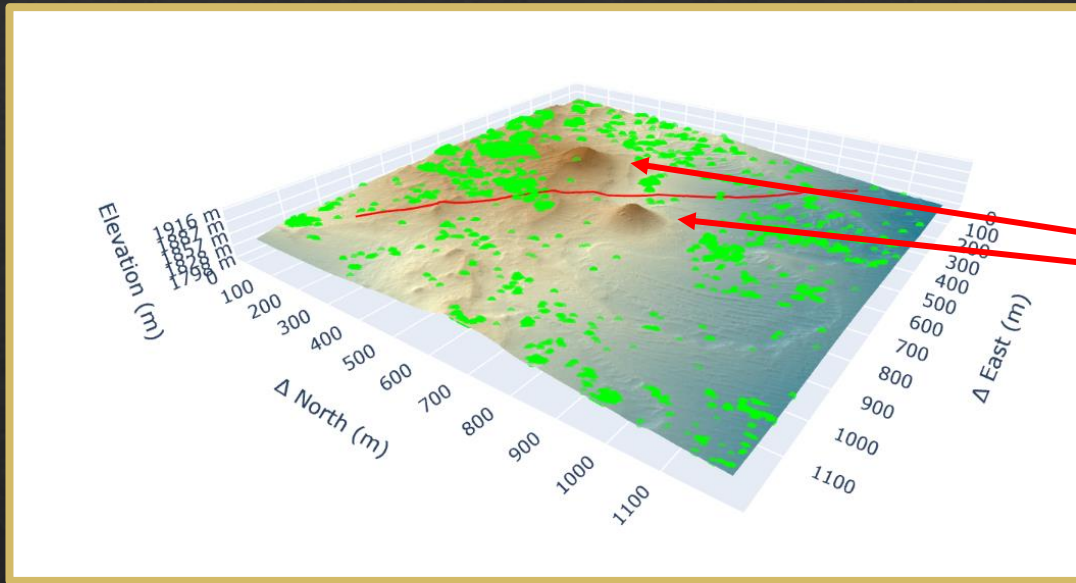


Topographic Profile

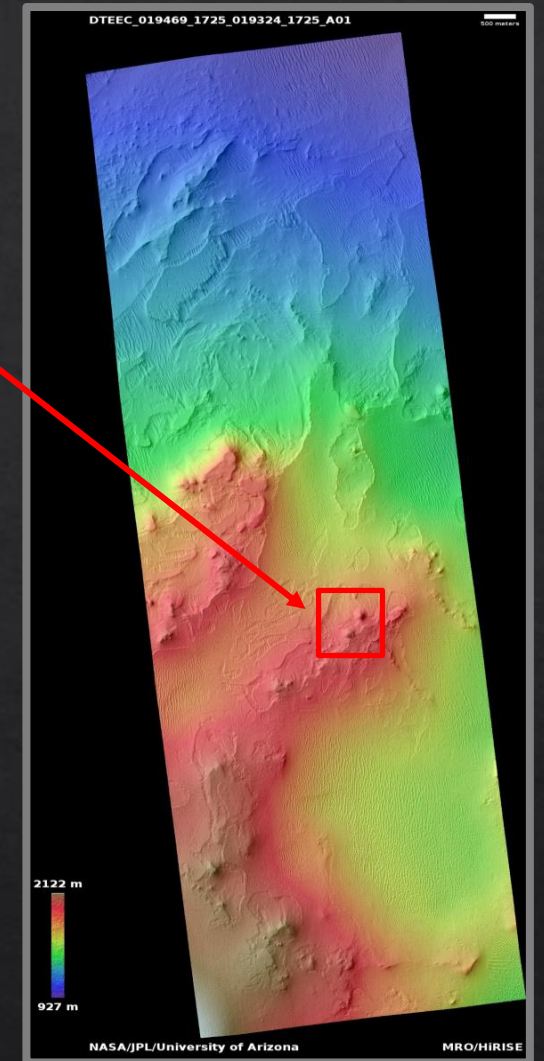
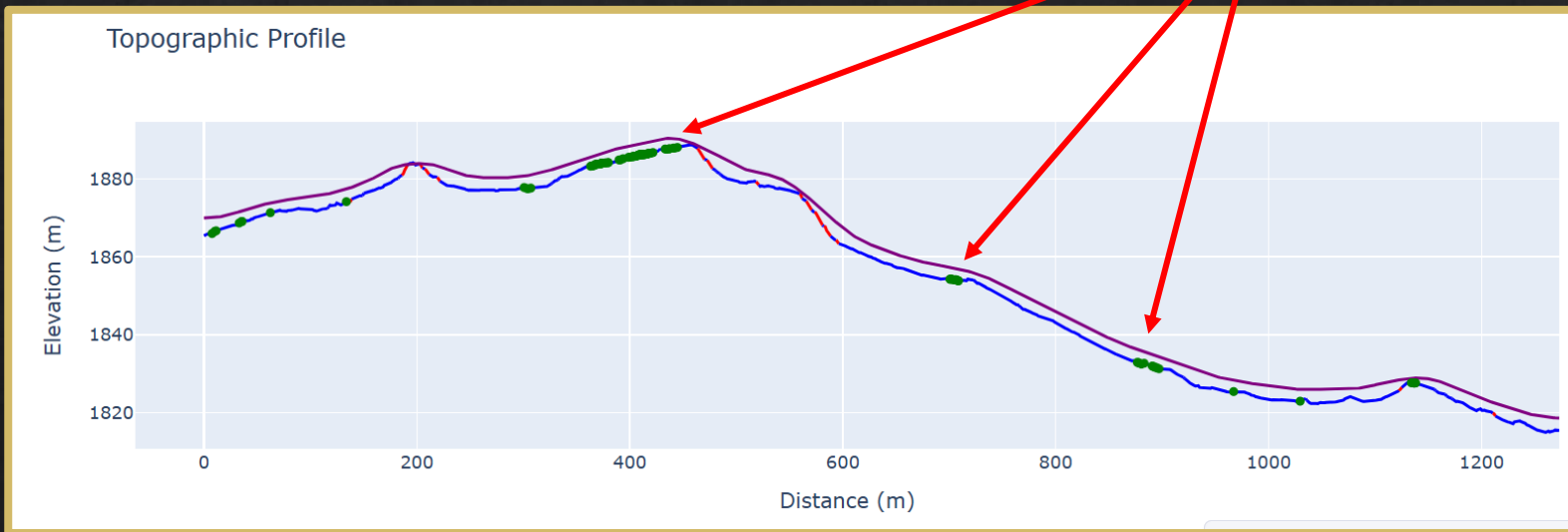


Layered Pit in Noctis Labyrinthus
7.7°S, 92.8°W

Longer Traversals



- Region of interest square expanded from 0.5km to 1.2km.
- Flight plan navigates around obstacles.
- Frequent airfields provide natural stopping points for recharge.



Light-Toned Outcrops in Noctis
Labyrinthus Region
13.4°S, 96.6°W

Key Findings

- ◆ MAESTRO enables flight planning for Mars rotorcraft, combining airfield detection, route analysis, and topographic profiles.
- ◆ Across diverse terrains, the presence of safe airfields varies dramatically, highlighting the need for planning tools.
- ◆ In regions like brain terrain, inclines, and craters, sustained flight and precise altitude control are essential due to limited or absent airfields.
- ◆ In flatter terrain, such as plains, dense airfield coverage supports flexible routing.
- ◆ By visualizing hazards and identifying safe airfields, MAESTRO reduces risk and expands the capabilities of scientific exploration by routing rotorcraft through previously inaccessible regions of Mars.

Future Work

- ◆ Overlay HiRISE imagery (25 cm/pixel) on DTMs (1 m/pixel) to maintain elevation information while adding finer terrain features such as smaller rocks and craters.
- ◆ Integrate traverse constraints (e.g. power, comms) and scientific targets to enable joint science/operations planning.
- ◆ Define high cost zones vs. low cost zones for traversal based on the number of available airfields to optimize routes
- ◆ Create multiple paths (suggested alternative routes) between airfields to reduce power consumption and fine elevation control integrating with Path Planning Optimizer

Recommendations

- ◆ Avoid relying on emergency landings in steep or irregular terrain—certain regions (e.g., brain terrain, crater walls) offer no viable airfields.
- ◆ Favor mission zones with dense, evenly distributed airfields, like flat plains or large troughs, to maximize flexibility and safety.
- ◆ Pre-select approaches into craters and sloped regions where MAESTRO identifies viable descent paths.
- ◆ Plan traversals around natural airfield clusters to create safe recharge points and backup landing sites.
- ◆ Incorporate MAESTRO in early-stage mission design to align science targets with terrain feasibility and path planning.
- ◆ Use MAESTRO terrain profiles to test controls for GPR-enabled flight (~ 4 m AGL).

Acknowledgements

This work is supported by the NASA-Berkeley Summer Scholars program including the Aerospace Engineering program at UC Berkeley. The author would like to thank NASA/JPL/University of Arizona for HiRISE data and the Mission Design Center at NASA Ames Research Center for mentorship and technical support.

